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**Seminar in Topics in Labor Economics and the Economics of
Education**

Did Remote Work Narrow the Motherhood Penalty?

**Evidence on the Intensive Margin from Israeli Labor Force
Survey Microdata, 2017–2023**

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Abstract

The post-pandemic shift to remote work is a persistent shock to the flexibility of employment, the job attribute mothers have historically paid most for. We ask whether the rise of work from home (WFH) narrowed the motherhood penalty in usual weekly hours among employed women in Israel. We use Central Bureau of Statistics (CBS) Labor Force Survey microdata on women aged 25–59 in 2017–2019 and 2021–2023 to estimate difference-in-differences (DiD) and triple-differences (DDD) models of usual weekly hours, with an Israeli-calibrated occupational WFH-exposure index as the third difference. Employed mothers’ hours did not change on average (0.23 hours), but they rose where the work can be done from home: the triple interaction is 3.22 hours per unit of exposure, or 1.71 hours a week between the mean exposures of the most and least teleworkable quartiles of occupations. Because exposure varies across only forty occupations, we report analytic, permutation and wild cluster bootstrap inference, and the gradient survives all three at the 10% level or better. The one exposure measure built entirely from pre-treatment information gives a weaker, insignificant estimate: the result requires the calibration’s reclassification of teaching, which Israeli schools kept in person. The gradient is largest for mothers of children under five, and fathers show the opposite gradient. The employment margin is estimated too imprecisely to support a comparable conclusion. The evidence indicates that remote work raises intensive-margin labor supply for mothers in teleworkable occupations specifically, rather than reducing the motherhood penalty across the board.

1 Introduction

The shift to remote work is a rare natural experiment in workplace flexibility. Unlike the acute disruption of 2020, work from home has persisted after the pandemic (Barrero, Bloom and Davis, 2021), and flexibility is the last major driver of the remaining gender pay gap (Goldin, 2014). Mothers pay for temporal and locational flexibility in lower wages or slower promotion. A shock that lowers its cost should therefore be detectable in mothers’ labor supply. For mothers already employed, it should show most directly in how many hours they work. This paper asks whether the post-COVID shift to remote work altered the motherhood penalty in *usual weekly work hours* among employed women in Israel, using Central Bureau of Statistics (CBS) Labor Force Survey microdata for 2017–2023.

One study has asked the question before, for the United States, where work from home reduced the motherhood penalty on employment and on income (Harrington and Kahn, 2025). We ask it of a different country and a different margin: how many hours the mothers who hold jobs supply. In doing so we provide, to our knowledge, the first estimate of the Israeli motherhood penalty in hours conditional on working. The occupational

WFH-exposure index the design rests on is calibrated to Israeli practice rather than imported unchanged. Israel is a natural setting for the intensive-margin question. It has the OECD’s highest total fertility rate, 2.87 children per woman in 2024 against an OECD average of 1.40 (OECD, 2025).

To identify the effect, we compare mothers and childless women across a pre-period of 2017–2019 and a post-period of 2021–2023, excluding 2020 (Section 3). A difference-in-differences on this contrast estimates the change in the hours penalty. A triple-differences model interacts this contrast with an occupation’s WFH exposure. It tests whether remote work is the *mechanism* by asking whether the change is larger in jobs that can actually be done from home. Its identifying assumption concerns how the motherhood gap *within* occupations would have evolved across exposure levels absent the shift.

Three findings emerge from this design. The first is a null: averaged over all employed women, the motherhood penalty in hours is no smaller after the pandemic than before it. The second is that the average conceals a gradient. In occupations whose work can be done from home, the hours gap between mothers and childless women closed by roughly three fifths of the penalty in those same occupations. In occupations tied to a workplace it did not close at all and, if anything, widened slightly. The third is that the gradient behaves as a flexibility mechanism should. It is largest for mothers of children under five, for whom the child-care time constraint binds most tightly, and it declines as the youngest child ages. Fathers in the same occupations show a gradient of the opposite sign. This rules out a shock common to everyone in teleworkable jobs, and it is consistent with a within-household reallocation of paid and care time, though the design does not identify one. Because the exposure score varies across only forty occupations, we report analytic, permutation and wild cluster bootstrap inference side by side. The strength of the statistical conclusion depends on which is used (Section 6.2). The estimate survives a battery of specification, selection and leave-one-occupation-out checks and a test of sorting across occupations. It does depend on the calibration of the exposure index in one identifiable way. US task content scores teaching as teleworkable, but Israeli schools kept it in person. Reclassifying teaching is what separates our estimate from the weaker, insignificant one given by the only fully pre-treatment measure (Section 6.3). We also estimate the employment margin, but for reasons of statistical power documented in Section 6.6 we treat it as secondary.

Section 2 reviews the literature, Sections 3 and 4 describe the data and the raw patterns, Section 5 sets out the design, Section 6 reports the results, and Sections 7–9 interpret them and state their limits.

2 Literature Review

Three questions organize the literature this paper draws on: why the motherhood penalty persists, what remote work does to the hours mothers supply, and how exposure to remote work can be measured, and its effect identified, when no one is assigned to it at random.

The penalty and the flexibility mechanism. After a first birth, women’s earnings fall sharply and persistently relative to men’s, through reduced hours, exit from employment and moves into family-friendly, lower-paying jobs. The size of the penalty tracks gender norms rather than family policy (Kleven, Landaïs and Søgaaard, 2019; Kleven et al., 2019). Because hours are one of those channels, a lower time cost of combining work and care can move the intensive margin without moving employment. The remaining gap has been located in the structure of jobs (Goldin, 2014). Many occupations pay disproportionately for long, contiguous, on-site hours, so mothers who cannot supply them pay a penalty out of proportion to the hours they forgo. Remote work lowers the cost of supplying a given number of hours, by removing the commute, letting hours be split around care and permitting a virtual presence in demanding professions. Both links have empirical support. Workers, and women in particular, give up wages for the option to work from home (Mas and Pallais, 2017). The skilled gender gap is concentrated where long hours are rewarded and narrows where substitutes for household time are cheap (Cortés and Pan, 2019). Lower visibility and slower advancement could cut the other way, so the sign is an empirical question. The place to look for it is the gradient across occupations by teleworkability, not the average.

What remote work does to hours, and when. Remote work has accounted for roughly a quarter of US paid workdays since 2021 (Barrero, Bloom and Davis, 2021, 2023), and the time it frees is split between paid work, care and leisure (Aksoy et al., 2023). Direct evidence on hours points both ways. Time worked rose under randomized home working and in the pandemic move home, in the latter case without output keeping pace (Bloom et al., 2015; Gibbs, Mengel and Siemroth, 2023). On remote days, however, part of the saved time goes to family and, for women, to household production, and teleworking mothers face more interruptions (Pabilonia and Vernon, 2022). The effect on gender gaps depends on how flexibility arrived. Under randomized or voluntary flexibility, household tasks shift toward men (Angelici and Profeta, 2024). Under the acute 2020 shock, with schools closed, women in WFH-feasible jobs absorbed the larger share of the resulting childcare, as was anticipated at the time (Alon et al., 2020; Adams-Prassl et al., 2020). Evidence on the post-acute period, once schools reopened and remote work became a job attribute rather than a lockdown condition, is much thinner. The one study of the motherhood penalty in that period (Harrington and Kahn, 2025) finds that WFH

raised US mothers' employment and income, with the employment effect concentrated in fields with high returns to hours and inflexible time demands. Hours are one channel of that income result, and the margin this paper observes directly.

Measuring exposure and identifying its effect. Remote work is chosen, and who selects into it differs sharply from what it does to a given worker (Emanuel and Harrington, 2024). A credible design therefore measures *exposure* from the content of jobs rather than from realized remote-work status. Two measures are in use. The *ex ante* measure asks whether an occupation's tasks can all be done at home, a criterion over a third of US employment meets (Dingel and Neiman, 2020). The *ex post* measure is the share of an occupation's workers who actually work from home. The first is immune to selection but identical across countries; the second reflects local practice but is itself a labor-market outcome. They diverge where institutions overrode task content. Teaching is the case in point: task content scores it as fully teleworkable, but Israeli schools kept it in person. Our index starts from the task-based classification mapped to two-digit ISCO-08 codes and calibrates it against realized Israeli remote work (Section 3.2), for which a pre-pandemic baseline exists (Madhala and Bental, 2021). A continuous exposure score also places the design in the continuous-treatment difference-in-differences family, where a single slope compares outcomes across exposure levels and need not weight those comparisons positively (Callaway, Goodman-Bacon and Sant'Anna, 2024). Section 5 takes up the two results that bear on reading our coefficient.

Israel, and the gap this paper fills. The Israeli evidence concerns other margins and other shocks. The gender wage gap is driven mainly by women's sorting into lower-paying firms, even within industry, with pay differences within firms contributing little (Buzaglo-Baris, 2023). An occupation-level exposure measure therefore captures only part of the relevant heterogeneity. Motherhood penalties in employment and earnings differ across ethno-religious groups (Budig, Kraus and Levanon, 2023), and the 2020 lockdown reallocated paid and unpaid work within households in gender-unequal ways (Yaish, Mandel and Kristal, 2021). To our knowledge, there is no published Israeli estimate of the motherhood penalty in hours conditional on working. Nor has any Israeli study asked whether remote work changed any margin of the penalty. This paper supplies both, with post-acute evidence from a high-fertility economy. It also shows, through the teaching occupations, that calibrating the exposure index to local practice can decide the result (Section 6.3).

3 Data and Institutional Context

3.1 Source and sample

The data are individual-level microdata from the Israeli Central Bureau of Statistics Labor Force Survey for the survey years 2017–2019 and 2021–2023. The analysis sample is women aged 25–59 (CBS age-group codes 3–7). A woman is a *mother* if she has any child aged 17 or younger in the household and childless otherwise. *Post* equals one for survey years 2021 and later. The comparison group therefore includes women whose children are grown and women who become mothers within the window (Section 8). The year 2020 is excluded: its arrangements were emergency improvisation, not settled practice, and no 2020 extract exists for this project.

The primary outcome is *usual weekly work hours*, recorded by the CBS in bins and assigned each bin’s median (coding in Appendix A). Hours are the only intensive-margin outcome the survey supports: the extract records how a wage is paid but not its amount. Hours are defined, in every year, for women who are employed *and* worked in the survey’s reference week, because the 2017 extract records zero hours differently (Appendix A). This restriction costs about 10% of each year’s employed sample and narrows the estimand to usual hours among women who were working (Section 8). The secondary outcome is a binary employment indicator (unemployed and not-in-labor-force are both zero).

All regressions include the same categorical controls: marital status, religion, age group, district of residence and highest educational attainment (six groups). Age enters as the CBS age-group code because the extract contains no continuous age. CBS survey weights are deliberately not applied in any outcome regression (Section 8). Replication code is in the project’s Git repository.¹

3.2 Constructing WFH exposure

Realized WFH status is itself a labor-market outcome, jointly determined with hours. The mechanism test therefore needs an occupation-level measure of *exposure* to remote work that the individual does not choose. Two ex-ante conditions select the primary measure. The score must be anchored on pre-period task content, so that it cannot be built from the post-2021 hours it is meant to explain. It must also not place in the treated tail occupations that Israeli institutions kept in person. We build three measures and report each.

External. The Dingel–Neiman teleworkability score (Dingel and Neiman, 2020) at the two-digit ISCO-08 level. It is pre-period by construction, but as a US task-based measure it misclassifies occupations where Israeli practice diverged, failing the second condition.

¹<https://github.com/NitzanZacharia/labor-econ-remote-work-analysis-hours.git>

Teaching scores 0.966, yet Israeli schools were fully in person throughout the 2022–23 survey years (Appendix B).

Realized. The realized Israeli WFH share by occupation, anchored to 2021. It is post-treatment by construction, failing the first condition.

Calibrated (primary). This index replaces the external score with the realized 2022–23 share only where the gap is large and statistically distinguishable from noise. Ten of the forty occupations are swapped, teaching and clerical support among them (rule, scores and fit in Appendix B). It alone meets both conditions, so it is primary; the other two are reported as bounds in Section 6.3. Because the swap test uses 2022–23 data, the calibrated measure is *partly post-treatment*. Section 6.3 tests whether the calibration produces the result.

A fourth measure serves three roles. Occupation is undefined for the non-employed, so the employment DDD uses a pre-period (2017–2019) shift-share index by demographic cell (sex, age group, education, district, marital status, religion and continent of birth; 8,884 cells). It averages the calibrated occupation score over the cell’s pre-period occupational composition and is therefore defined for employed and non-employed women alike. The same index stratifies the selection correction of Section 5.4. Because a woman’s post-2021 occupation cannot move it, it also bounds occupational sorting in Section 6.3. Its dispersion is a third of the occupation-level index’s, so coefficients on the two are compared per standard deviation, not per unit.

4 Descriptive Statistics

4.1 The sample in summary

Table 1 describes the sample on either side of the treatment/control split. Mothers work *fewer* hours than childless women when they work (38.68 versus 40.06) while being slightly *more* likely to be employed (0.78 versus 0.76), so the penalty’s level is visible on the intensive margin; the table cannot show whether it changed after 2021. The two groups are imbalanced in ways the controls must absorb: mothers are the better-educated group (48% with an academic degree against 40%), overwhelmingly married (85% against 47%), and about 0.44 of an age-group code younger, an imbalance that tracks WFH exposure (Appendix Table A4).

4.2 The raw gap and its exposure gradient

Mothers work 38.54 hours per week before 2021 and 38.84 after, a gain of 0.31 hours; childless women move from 40.04 to 40.08. The difference between those two changes, 0.27 hours, is the difference-in-differences of equation (1) without a single control, and the regression DiD of Table 2, 0.228 (SE 0.181), is within four hundredths of an hour of it.

Table 1: Summary statistics: women aged 25–59, by mother status

	Childless women	Mothers	Difference (SE)
<i>Panel A: outcomes</i>			
Observations	133,646	239,095	
Employment rate	0.7574	0.7828	0.0254 (0.0031)
Usual weekly hours (ref.-week workers), mean	40.06	38.68	−1.38 (0.09)
standard deviation	[12.12]	[11.62]	
observations with usual hours	93,982	164,193	
<i>Panel B: characteristics</i>			
Age-group code (CBS 3–7), mean	5.40	4.96	−0.44 (0.01)
Arab (%)	17.73	17.78	0.05 (0.33)
<i>Marital status (%)</i>			
Married	46.55	84.57	38.02 (0.35)
Married, living separately	3.58	0.84	−2.74 (0.11)
Divorced	13.13	7.70	−5.42 (0.24)
Widowed	2.75	0.99	−1.76 (0.11)
Single, never married	34.00	5.90	−28.10 (0.31)
<i>Religion of household head (%)</i>			
Jewish	73.08	79.50	6.42 (0.36)
Christian	7.12	2.23	−4.89 (0.18)
Muslim	13.78	14.89	1.11 (0.30)
Druze	1.88	1.64	−0.24 (0.12)
Other	4.15	1.75	−2.40 (0.14)
<i>District of residence (%)</i>			
Jerusalem	8.76	12.31	3.56 (0.26)
North	17.52	16.58	−0.94 (0.32)
Haifa	14.38	11.92	−2.45 (0.28)
Center	23.99	26.52	2.53 (0.36)
Tel Aviv	21.55	14.90	−6.65 (0.33)
South	11.66	12.37	0.72 (0.27)
Judea and Samaria	2.15	5.39	3.24 (0.15)
<i>Education (%)</i>			
Below high school	11.37	6.63	−4.74 (0.23)
High school, no matriculation	12.70	11.95	−0.74 (0.24)
Matriculation (Bagrut)	22.73	18.82	−3.90 (0.29)
Post-secondary, non-academic	11.50	13.68	2.18 (0.24)
Academic degree (BA/MA/PhD)	40.24	47.97	7.73 (0.39)
Other / no certificate	1.47	0.95	−0.52 (0.07)

Israeli CBS Labor Force Survey, survey years 2017–2019 and 2021–2023, women aged 25–59; a mother has any child aged 17 or younger in the household. Weekly hours are conditional on working in the reference week, with the standard deviation in brackets and their own observation count. The difference column is mothers minus childless women, with a standard error clustered by individual. All figures are *unweighted*. The Arab share is the CBS population-group code used for the subgroup analysis; religion is a separate item recorded for the household head. Marital status, religion, district, age group and education are the five controls in every regression.

Mothers remain well below childless women in both periods, and the raw gap widened by 0.3 hours over the three pre-period years, within sampling error (Appendix Figure A1). Figure 1 computes the same difference-in-differences separately within each quartile of occupational WFH exposure. The bottom three quartiles return -0.32 , 0.27 and -0.02 hours, none distinguishable from zero; the top quartile returns 1.55 hours (95% CI $[0.74, 2.36]$). The near-zero average is therefore a weighted average of three nulls and one substantial positive difference. The figure is a description rather than an independent causal test: occupational composition and time-varying confounders can differ across quartiles, and the regressions of Section 6 address that. The response is not linear in exposure: the bottom three quartiles span exposure values of 0.00 to 0.30 and the top quartile 0.36 to 0.75, so most of the regressor’s dispersion, and the identifying contrast, sits in one tail.

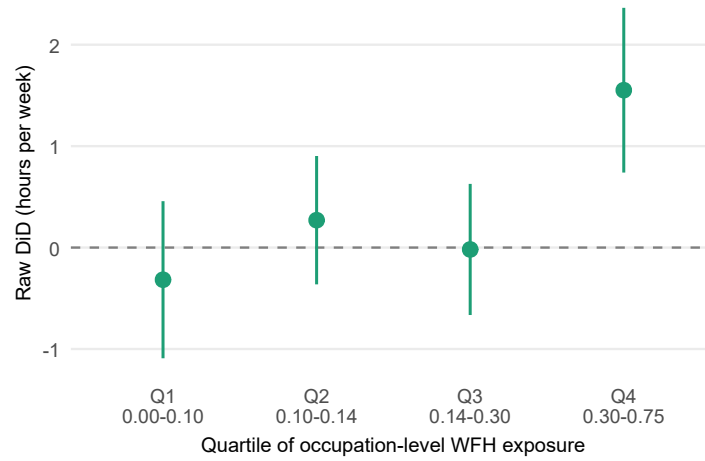


Figure 1: Raw hours difference-in-differences by quartile of WFH exposure

Notes: Women aged 25–59 who worked in the reference week and carry a matched occupation code. Each point is the raw difference-in-differences within one quartile of the occupation-level calibrated exposure measure, from unweighted cell means with no controls; bars are 95% confidence intervals clustered by individual. Breakpoints are computed on pre-period rows and applied to all years; each quartile’s exposure range is printed beneath its tick. The 1.55 hours in the top quartile is that quartile’s own difference-in-differences; the raw contrast between the top and bottom quartiles is $1.55 - (-0.32) = 1.87$ hours.

5 Empirical Strategy

5.1 Difference-in-differences

For outcome Y_{it} (weekly hours conditional on working that week, or employment) we estimate

$$Y_{it} = \beta_0 + \beta_1 \text{Mother}_i + \beta_2 \text{Post}_t + \beta_3 (\text{Mother}_i \times \text{Post}_t) + X'_{it} \gamma + \varepsilon_{it}, \quad (1)$$

where X_{it} is the control vector of Section 3. β_3 is the differential change in the outcome for mothers relative to childless women after the shift to remote work.

5.2 Triple differences

The flexibility framework (Goldin, 2014) yields one testable hypothesis: after 2021, mothers' usual hours conditional on working rise relative to childless women's in proportion to their occupation's WFH exposure. Where the work cannot be done from home, there is no response. The exposure gradient, not the average change, is the object of interest, so we interact the DiD with occupational WFH exposure:

$$\begin{aligned} Y_{it} = & \alpha + \delta_1 \text{Mother}_i + \delta_2 \text{Post}_t + \delta_3 \text{WFH_Exposure}_{j(i)} + \delta_4 (\text{Mother}_i \times \text{Post}_t) \\ & + \delta_5 (\text{Mother}_i \times \text{WFH_Exposure}_{j(i)}) + \delta_6 (\text{Post}_t \times \text{WFH_Exposure}_{j(i)}) \\ & + \delta_7 (\text{Mother}_i \times \text{Post}_t \times \text{WFH_Exposure}_{j(i)}) + X'_{it} \gamma + \varepsilon_{it}, \end{aligned} \quad (2)$$

where $j(i)$ is individual i 's occupation. δ_7 asks whether the post-shift change for mothers relative to non-mothers is larger in more exposed occupations. δ_6 absorbs any post-2021 change in teleworkable occupations common to mothers and childless women, and δ_5 the pre-existing level of the penalty across occupations. δ_4 is the post-shift change for mothers at *zero* exposure, not an average effect. The DDD requires a single parallel-trends assumption (Olden and Møen, 2022): that the mother/non-mother gap in high- versus low-exposure jobs would have evolved in parallel absent the shift to remote work. Insignificant pre-period coefficients are evidence about that assumption, not a verification of it. The hours DDD uses the calibrated occupation score, WFH_Exposure_j , on employed women matched to a two-digit occupation. The employment DDD uses the cell-based index on all women and adds a $\text{Mother} \times \text{age-group}$ interaction for the pre-period age imbalance.

Equation (2) makes each lower-order occupation-level difference linear in exposure, an unnecessary restriction with forty occupations. A single slope also compares effects *across* exposure levels. That requires the stronger condition that occupations at different exposure levels would have followed the same path under any common level of exposure, and the slope is a weighted average of such comparisons whose weights need not be positive

(Callaway, Goodman-Bacon and Sant’Anna, 2024). We therefore estimate two relaxations alongside it. The *saturated* specification replaces every lower-order term with fixed effects,

$$Y_{it} = \delta_7(\text{Mother}_i \times \text{Post}_t \times \text{WFH_Exposure}_{j(i)}) + X'_{it}\gamma + \lambda_{j(i),t} + \mu_{j(i),\text{Mother}_i} + \nu_{\text{Mother}_i,t} + \varepsilon_{it}, \quad (3)$$

where λ_{jt} , $\mu_{j,m}$ and $\nu_{m,t}$ are occupation-by-year, occupation-by-mother and mother-by-year effects, which absorb every lower-order term of equation (2) and any occupation-specific year shock. The *binned* specification replaces $\text{WFH_Exposure}_{j(i)}$ with indicators for the occupation’s pre-period exposure quartile, the bins of Figure 1. The top-versus-bottom contrast then carries a standard error.

5.3 Inference

The survey is a rotating panel in which most respondents appear about four times, so equation (1) and every descriptive interval are clustered by individual (Appendix A gives the panel counts). The employment DDD is clustered on the age-by-education-by-district cell (about 210 clusters). For the hours DDD the regressor is constant within a two-digit occupation, so standard errors are clustered at that level, leaving forty clusters. The reported p -values use a t reference distribution with $G - 1 = 39$ degrees of freedom and a cluster-count adjustment to the variance.

Forty is few enough that we do not rest on that approximation. For the headline coefficient and the pre-period coefficients of the triple-difference event study we also report wild cluster bootstrap p -values, with Rademacher weights, the null imposed and 9,999 draws (Roodman et al., 2019; Fischer and Roodman, 2021). The same bootstrap is applied to every occupation-clustered robustness and subgroup refit and printed beside its analytic p -value, since a star from a test that over-rejects is weaker evidence than it looks. The joint pre-period test remains the analytic Wald F , since a joint bootstrap test is not available in the implementation used. Section 5.5 adds a permutation test that needs no asymptotic approximation. The three procedures answer related but different questions, and where they disagree the disagreement is part of the evidence. For each DDD we also report the minimum detectable effect, $\text{MDE} = \text{SE}(t_{0.975} + t_{0.80})$, which is $2.87 \times \text{SE}$ under the t with 39 degrees of freedom and $2.80 \times \text{SE}$ under the normal. A null can then be read as “genuinely null” or “underpowered.”

5.4 Selection into employment: Lee bounds

Hours are observed only for women who are working, and working is itself an outcome of the same treatment. We therefore report trimming bounds (Lee, 2009). The share of post-period mothers observed with hours in excess of a parallel-trends counterfactual is trimmed from that cell’s hours distribution: from the top for a lower bound, from the

bottom for an upper bound. The model is refit on each trimmed sample, and we report the Imbens–Manski interval for the identified set (Imbens and Manski, 2004). For the DDD the trimming is done within quartiles of the cell-based index, which is defined for the non-employed. The bounds address excess selection under monotonicity; they cannot address selection that violates it, nor differences in absence unrelated to excess selection. Appendix C gives the construction and its assumptions.

5.5 Diagnostics, falsification, and the fathers’ comparison

We estimate event-study versions of equation (1) for both outcomes, with Mother $\times$ year interactions and 2019 as the reference year, and test the pre-period coefficients jointly. That test bears on the difference-in-differences assumption, not the triple-differences one. We therefore also estimate the event-study version of equation (2), with Post replaced by survey-year interactions at every level, and apply the same joint test to its pre-period coefficients.

The falsification test is a permutation test on the exposure score. Under the sharp null that teleworkability is unrelated to the post-2021 change in the motherhood gap, the forty occupation scores are exchangeable. We therefore reassign them at random across the forty occupation codes 999 times, holding every woman’s occupation and the mother/period structure fixed, refit equation (2), and record the cluster-robust t -statistic of the triple interaction. The permutation p -value is the share of draws, counting the observed one, whose $|t|$ reaches the observed value (Phipson and Smyth, 2010). Occupation is observed in the same survey as the outcome, so remote work could have drawn mothers into teleworkable occupations after 2021. We check this with a difference-in-differences that takes the exposure score itself as the outcome, and bound the channel by re-estimating the hours DDD on the pre-period cell index of Section 3.2.

The remaining checks, collected in Table 3, re-estimate the hours DDD varying in turn the exposure measure, the age control, the sample, the outcome coding and the inference procedure. We also re-estimate it by the age of the youngest child, separately for Jewish and Arab women, and on the male subsample with “mother” read as “father.” The last is a comparison, not a placebo. Fathers in teleworkable occupations are exposed to remote work exactly as mothers are, so a negative gradient would be evidence on household reallocation rather than on a confound. Two-sample z -tests compare coefficients across disjoint subsamples.

6 Results

Three findings organize this section. The average motherhood penalty in weekly hours did not change after the shift to remote work. It did narrow where the work can be done from

home: the post-2021 change in the mother/childless gap rises with an occupation’s WFH exposure and is concentrated in the most teleworkable quartile. That gradient survives the design’s checks on selection, sorting, calibration and specification, is largest for mothers of young children, and is reversed in sign for fathers. The employment margin is too imprecisely estimated to support a comparable conclusion. Tables 2–5 and Figure 2 carry the estimates; the text reports what they show. Significance is denoted *** $p < 0.001$, ** $p < 0.01$, * $p < 0.05$, · $p < 0.1$, with standard errors in parentheses. Stars follow the analytic occupation-clustered t -test, which over-rejects with forty clusters (Section 5.3); Tables 3–5 therefore also print a wild cluster bootstrap p -value beside every occupation-clustered triple interaction.

6.1 No average change in the hours penalty

Column (1) of Table 2 estimates equation (1). Mothers work 1.739 fewer hours per week than childless women, and that penalty did not measurably change after 2021. The 95% confidence interval of Mother $\times$ Post, $[-0.13, 0.58]$, excludes average gains above about half an hour a week, and the design could have detected an effect of about 0.51 hours. Panel (a) of Figure 2 supports the comparison: the pre-period coefficients are indistinguishable from zero and pass a joint Wald test, and the post-period path is flat apart from a larger 2023 coefficient. Survey-year effects in place of the single Post dummy, and the selection bounds of Appendix C, leave the reading unchanged. Nothing that follows rests on this average.

6.2 Hours rose where the work can be done from home

Column (2) of Table 2 interacts the DiD with the calibrated occupation-level exposure index. The triple interaction is 3.224** (SE 1.022). Among working mothers, those in more WFH-exposed occupations saw a larger post-2021 increase in weekly hours relative to childless women in the same occupations. Between the mean exposures of the bottom and top quartiles of Figure 1 that is 1.71 hours, the pattern the figure shows in raw means. The implied change at zero exposure is negative and insignificant, so mothers in jobs that cannot be done from home gained nothing. What changed after 2021 is not mothers’ hours in general but mothers’ hours where remote work was possible.

Columns (3) and (4) relax the functional form and tell the same story. With the three sets of fixed effects of equation (3) absorbing every lower-order term, the triple interaction falls by a tenth and remains significant at the 1% level. With quartile indicators in place of the score, the second and third quartiles differ from the least teleworkable one by positive but insignificant amounts, and the top quartile by a significant amount close to the pooled model’s 1.71-hour contrast. The response is a step at the top of the exposure distribution more than a smooth gradient. That quartile holds six of the forty occupations, but the pooled and saturated columns use all forty and agree with it, and no single occupation

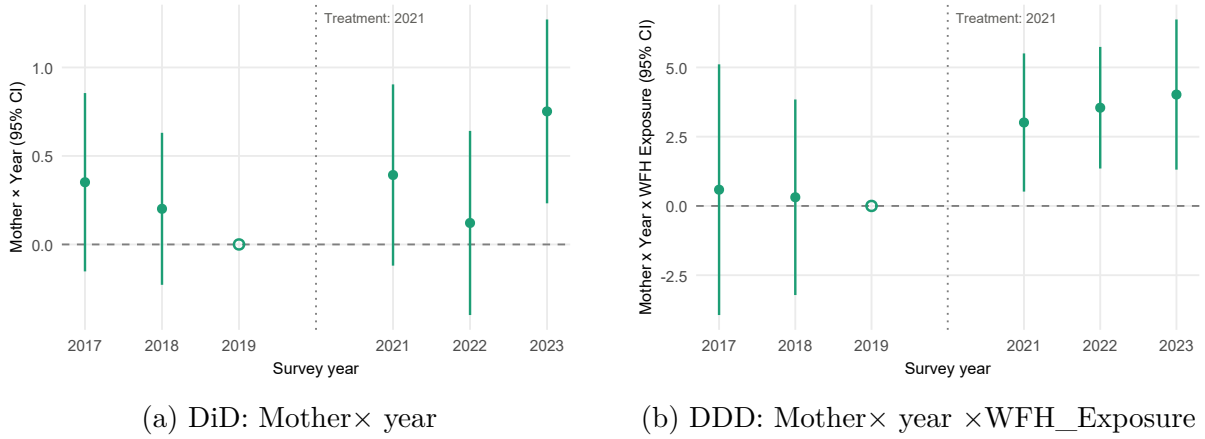


Figure 2: Event studies for weekly hours, reference year 2019

Notes: Vertical bars are 95% confidence intervals; the hollow point at 2019 is the omitted reference year, and the gap between 2019 and 2021 is the excluded 2020 survey year. Panel (a): women working in the reference week ($N = 251,857$), standard controls, standard errors clustered by individual. Coefficients (SE): 2017, 0.352 (0.257); 2018, 0.201 (0.219); 2021, 0.392 (0.261); 2022, 0.121 (0.265); 2023, 0.752** (0.265). Joint test on 2017–2018: $F(2, 63,198) = 0.949$, $p = 0.387$. Panel (b): sample, controls, exposure and clustering as in column (2) of Table 2 ($N = 246,326$, 40 clusters), with the year main effects and both two-way year interactions alongside the plotted triple interaction. Coefficients (SE), hours per unit of exposure: 2017, 0.587 (2.237); 2018, 0.313 (1.745); 2021, 3.011* (1.233); 2022, 3.545** (1.085); 2023, 4.020** (1.340). Joint test on 2017–2018: $F(2, 39) = 0.041$, $p = 0.960$; wild cluster bootstrap p -values (Rademacher, 9,999 draws) 0.838 for 2017, 0.884 for 2018 and 0.852 for their sum.

drives it (Section 6.4).

Panel (b) of Figure 2 is the event-study version of equation (2), the pre-trend test the triple difference itself requires (Section 5.5). The pre-period coefficients are near zero and jointly insignificant on both the Wald test and the wild cluster bootstrap. The post-period path steps up in 2021 and rises through 2023, every year straddling the pooled estimate. The test has limited power. With forty clusters each pre-period interval contains values as large as the estimate itself. It shows no sign of a violation; it does not rule one out.

How precisely the gradient is estimated depends on the inference procedure, and we report three. The analytic occupation-clustered standard error on a t_{39} reference gives $p = 0.003$. The two procedures built for forty clusters are less emphatic. The permutation test, which reassigns the forty scores across occupations, gives $p = 0.032$ (Appendix Figure A2). The wild cluster bootstrap gives $p = 0.056$, with a 95% confidence set of $[-0.26, 5.84]$ that includes zero. We report all three rather than choosing among them.

6.3 Selection, sorting, and the calibrated regressor

Three features of the design could produce an exposure gradient with no change in mothers' labor supply. They are selection into work, sorting of mothers across occupations, and an index calibrated on post-period data. Table 3 reports the tests.

Table 2: Intensive-margin DiD and DDD: weekly hours, reference-week workers

	(1) DiD	(2) DDD	(3) DDD, saturated	(4) DDD, quartiles
Mother $\times$ Post $\times$ WFH_Exposure		3.224** (1.022)	2.876** (0.950)	
Mother $\times$ Post $\times$ Q2 (vs. Q1)				0.753 (0.483)
Mother $\times$ Post $\times$ Q3 (vs. Q1)				0.600 (0.485)
Mother $\times$ Post $\times$ Q4 (vs. Q1)				1.844** (0.595)
Mother $\times$ Post	0.228 (0.181)	−0.539 (0.336)		−0.472 (0.346)
Mother $\times$ WFH_Exposure		−1.295 (1.053)		
Post $\times$ WFH_Exposure		−0.548 (0.810)		
WFH_Exposure		1.128 (3.912)		
Mother	−1.739*** (0.144)	−1.449*** (0.351)		−1.527** (0.452)
Post	−0.057 (0.148)	0.053 (0.273)		−0.130 (0.367)
Constant	32.355*** (0.318)	32.115*** (1.540)		31.581*** (2.051)
MDE for triple interaction (t_{39})		2.938		
SD of exposure, estimation sample		0.199		
Triple interaction per SD of exposure		0.640		
Occupations in Q1 / Q2 / Q3 / Q4				16 / 7 / 11 / 6
Fixed effects	None	None	Three sets	None
Observations	251,857	246,326	246,324	246,326
R^2	0.040	0.041	0.135	0.048
Clustering	individual	occupation (40)	occupation (40)	occupation (40)

Dependent variable: usual weekly work hours, women working in the reference week; standard controls in every column. Columns (2)–(4) use the calibrated occupation-level exposure index on column (1)’s sample less the rows with no matched two-digit occupation. Column (3) is equation (3), whose three sets of fixed effects (occupation $\times$ year, occupation $\times$ mother, mother $\times$ year) absorb every lower-order term. Column (4) replaces the score with indicators for its pre-period quartile (Figure 1’s bins); its lower-order quartile terms are estimated but not shown, and the number of occupations per bin is listed because occupation is the cluster, counted on the full 2017–2023 estimation sample (Appendix Table A4 counts pre-period rows only, so its bottom quartile shows one occupation fewer). Columns (2)–(4) use a t reference distribution with 39 degrees of freedom (Section 5.3), which the MDE also uses ($\alpha = 0.05$, power 0.80, multiplier 2.87); the wild-bootstrap and permutation p -values for column (2)’s triple interaction are in Table 3. Column (2)’s Mother $\times$ Post is the effect at *zero* exposure, not an average effect. Coefficients and standard errors to three decimals throughout. No coefficient is dropped for collinearity in any column.

Selection into work. Trimming the excess selection among post-period mothers within each quartile of the cell-based index (Appendix C) barely moves the estimate in either direction, and the Imbens–Manski 95% interval for the identified set excludes zero.

Sorting across occupations. If remote work drew mothers into teleworkable occupations after 2021, the occupation score would itself respond. It does not. As a dependent variable it gives a Mother $\times$ Post under one hundredth of its standard deviation, the probability of being in the top quartile does not move, and the event-study version is flat in every year. The hours DDD on the pre-period demographic-cell index, which a woman’s later occupation cannot move, is positive but imprecise. That index is a far coarser measure of a woman’s own teleworkability, so the sorting DiD is the informative test.

Calibration. The calibrated index draws on realized 2022–23 WFH shares, so the regressor could be absorbing post-period outcomes. The external Dingel–Neiman index (Dingel and Neiman, 2020), the only measure built entirely from pre-treatment information, gives a positive but insignificant gradient about a fifth the size of the headline, the weakest of the three exposure measures. The realized 2021-anchored index gives a larger, noisier estimate. Three checks hold the sample at all forty occupations. Calibrating on men only, so that no woman in the regression contributes to her own regressor, gives a gradient above the headline. Keeping the external score but letting the ten reclassified occupations carry their own Mother $\times$ Post change restores a gradient in the external score at nearly three quarters of the headline. The occupations the external index places in the treated tail did not respond. Reclassifying teaching alone, with the other thirty-nine occupations at their external scores, recovers a significant gradient at three fifths of the headline. Dropping the ten reclassified occupations instead, so that the two indices coincide on the thirty that remain, gives one above it. Moving the calibration’s swap threshold across its full range leaves the estimate positive throughout, from the setting that swaps only teaching and clerical support to the one that swaps fifteen occupations (Appendix Table A2). The result therefore requires reclassifying teaching. That reclassification rests on an institutional fact rather than on any woman’s hours: Israeli schools were fully in person in 2022–23 (Appendix B). The calibration cannot be made into a pre-treatment measure. A reader who weights the external index above the others will read the headline as directionally supported rather than established.

6.4 Robustness of the gradient

Two of the remaining checks in Table 3 matter, and both concern balance. Mothers and childless women differ in age, education and marital status, and the gap itself varies with exposure (Appendix Table A4). Such a gap is exactly what the triple interaction is

identified off. A Mother $\times$ age-group interaction leaves the estimate essentially unchanged. Reweighting to balance pre-period age groups attenuates it by about 19%. The reweighted estimate is still significant at the 5% level on the analytic test but now below the design’s minimum detectable effect.

Marital status is the imbalance the age checks do not touch. Most mothers are married and fewer than half of childless women are (Table 1), and the gap widens with exposure. Restricting the sample to married women, or adding marital status $\times$ Post $\times$ WFH_Exposure, leaves the gradient positive but without conventional significance. Part of the headline may therefore come from the comparison group. Never-married childless women, a larger share of it at high exposure, may have worked fewer hours after 2021. Neither row is a clean test. The interaction is estimated largely off mothers themselves, most of whom are married, so it partly over-controls. The married subsample trades a marital confound for a life-stage one, since married childless women are older and often have grown children. Both are the bounded version of a weakness the design cannot remove.

Everything else stays close to the headline. The last quarter of the 2023 survey year includes the start of the October 2023 war (Section 8). Excluding that year moves the estimate by less than one standard error and leaves the plain hours DiD a null. Dropping any one of the forty occupations does the same, and every refit stays significant at the 5% level on the analytic test (Appendix Figure A3). Across every specification in the table that keeps a calibrated index, the triple interaction lies between 2.2 and 4.3 hours. The wild cluster bootstrap column of Table 3 tempers the stars. Across the same rows its p -values run from 0.02 (unswapped occupations, men-only calibration) to 0.13 (marital interaction).

Nor does the outcome coding drive it. Hours are a bin-median approximation and the effect is smaller than one bin width, but dropping the two imputed irregular-hours codes changes little. Because 35 and 40 hours are exact bin edges, the same specification can be estimated as a linear probability model on either threshold. On both, mothers’ probability of crossing it rose after 2021 by about 2 percentage points per standard deviation of exposure relative to childless women’s. The open 60-plus bin is assigned the centre of its range, a coding choice rather than a measurement. Recoding it to 60 lowers the estimate by a fifth. A linear probability model for that bin is itself significantly positive: mothers’ share of the longest-hours bin rose where the work can be done from home. The effect therefore appears at the full-time boundary and at the top of the distribution. Its size depends in part on the value assigned to the open bin.

Table 3: Hours DDD: sensitivity of Mother $\times$ Post $\times$ WFH_Exposure

Specification	Coefficient	SE	p	p (boot)	N	Clusters
<i>Exposure measure</i>						
Calibrated (primary)	3.224**	1.022	0.0031	0.0558	246,326	40
External (Dingel–Neiman)	0.672	0.884	0.4517	0.5780	246,326	40
Realized (2021 anchor)	5.195*	2.363	0.0346	0.0278	246,037	36
Calibrated on men only (12 of 40 swapped)	3.869***	0.979	0.0003	0.0244	246,326	40
External index, teaching (ISCO 23) alone swapped	1.934***	0.519	0.0006	0.0020	246,326	40
External index, swapped-occupation terms added	2.339	1.352	0.0915	0.2074	246,326	40
Mother $\times$ Post $\times$ Swapped (same model)	−1.347	0.926	0.1537	0.3041	246,326	40
<i>Age and marital balance</i>						
Age-interacted (+ Mother $\times$ age group)	3.205**	1.055	0.0042	0.0541	246,326	40
Reweightd (pre-period age-group raking)	2.597*	1.050	0.0178	0.0539	242,552	40
Married women only	2.743	1.512	0.0775	0.1119	173,934	39
Marital status $\times$ Post $\times$ WFH_Exposure added	2.191	1.273	0.0932	0.1254	246,326	40
<i>Sample</i>						
Unswapped occupations only	4.309**	1.192	0.0011	0.0201	132,046	30
Excluding survey year 2023	2.979*	1.170	0.0151	0.0739	208,833	39
Absentees at usual hours, 2018–2023	3.076***	0.861	0.0010	0.0694	224,567	40
Leave-one-occupation-out (40 refits): range	[2.415, 3.580]		[0.0007, 0.0167]			39
<i>Occupational sorting</i>						
Pre-period cell exposure as regressor, per SD	0.139	0.159	0.3823		247,931	245
Exposure score as outcome: Mother $\times$ Post	0.001	0.003	0.6996		246,326	individual
Top-quartile indicator as outcome: Mother $\times$ Post	0.004	0.007	0.5433		246,326	individual
<i>Outcome coding</i>						
Irregular-hours codes dropped, not imputed	3.056**	1.059	0.0064	0.0573	232,785	40
Full-time indicator (≥ 35 hours)	0.096**	0.029	0.0020	0.0466	246,326	40
Long-hours indicator (≥ 40 hours)	0.107**	0.038	0.0075	0.0529	246,326	40
Top bin (60+) recoded to 60 hours	2.589**	0.917	0.0074	0.0625	246,326	40
Top-bin indicator (60+ hours)	0.034***	0.009	0.0005	0.0937	246,326	40
<i>Inference on the primary estimate</i>						
Two-way clustering (ind., occ.)	3.224**	1.022	0.0031		246,326	40 $\times$ ind.
Wild cluster bootstrap ($B = 9,999$)	3.224	1.022	0.0558		246,326	40
Permutation test (999 reassignments)	3.224		0.0320		246,326	40

Unless a row says otherwise: the triple interaction Mother $\times$ Post $\times$ WFH_Exposure from equation (2), women working in the reference week, standard controls, calibrated exposure, clustered by occupation with the cluster count in the last column, p the analytic $t(G - 1)$ p -value and p (boot) the wild cluster bootstrap p -value of the same coefficient (Rademacher weights, null imposed, 9,999 draws), blank where the row is not a one-way occupation-clustered refit of a single model (the two-way-clustered, leave-one-out, sorting and cell-index rows). The men-only row calibrates the index on men’s realized 2022–23 WFH shares; the teaching-only row moves ISCO 23 alone from its external score to its realized share and leaves the other thirty-nine at the external score; the married-only row drops marital status from the controls; the marital-interacted row adds marital status $\times$ Post $\times$ WFH_Exposure and its two-way terms; the swapped-occupation rows keep the external index as regressor on all forty occupations and add a Swapped indicator for the ten calibrated occupations with the full Mother $\times$ Post structure, showing both triple interactions from that one model; the unswapped-occupations row drops those ten; the absentees row adds employed women absent from work in the reference week at their reported usual hours, on 2018–2023 only, since the 2017 file recorded absentees’ hours as zero. The leave-one-occupation-out row gives the range of the forty refits with one occupation removed (Appendix Figure A3) and the range of their p -values. Under *Occupational sorting*, the first row is the hours DDD with the pre-period demographic-cell index of Section 3.2 as regressor, reported per standard deviation of that index and clustered on the demographic cell; the next two are differences-in-differences with the occupation-level score, and an indicator for its top quartile, as the dependent variable, clustered by individual. The two indicator outcomes under *Outcome coding* are linear probability models on bin-crossing outcomes; the top-bin rows recode the open 60-plus bin, assigned 78.5 hours in the main coding, to 60 hours, and estimate a linear probability model for being in that bin. The wild-bootstrap row reports the analytic coefficient and standard error with the bootstrap p -value; the permutation row reports the permutation p -value, which has no standard error.

6.5 Who responded: child age, population group, and fathers

If remote work relaxes a time constraint that binds through child care, the response should be largest for mothers of young children and smallest for mothers of teenagers. It is. Table 4 re-estimates the DiD and DDD for mothers whose youngest child is aged 0–4, 5–9, 10–14 or 15–17, each against the same childless women. The average effect is a null in every group. The exposure gradient declines monotonically with the child’s age; for mothers of 15–17-year-olds it is half the size of the under-fives’. The four regressions share a control group and the same forty clusters, so their intervals overlap and are not independent, and on the wild cluster bootstrap none of the four is significant at the 5% level. The pattern is a gradient consistent with the mechanism rather than a set of statistically separated effects, and the estimate is not carried by mothers of teenagers.

Table 5 splits the sample by population group and re-estimates the models on men. Jewish women reproduce the pooled pattern. Arab women’s gradient is nominally twice as large but statistically indistinguishable from it on a subsample nine times smaller: imprecise rather than a documented heterogeneous effect.

Re-estimating the DDD on men, so that fathers are compared with childless men in the same occupations, gives a gradient of the opposite sign. On its own the fathers’ coefficient is marginal: it is not significant at the 5% level on either test. It is nonetheless distinguishable from the mothers’, since a two-sample test on the difference between the two gradients rejects equality. Fathers’ average hours also fell relative to childless men’s, where mothers’ average did not move, and the women-versus-men difference in the DiD is itself significant. The comparison is not a placebo (Section 5.5), so a null was never the prediction. What it shows is that a macro trend common to all workers in teleworkable jobs cannot produce a positive gradient for mothers beside a negative one for fathers. Opposite gradients are also what within-household reallocation would look like if mothers’ additional paid hours were matched by fathers’ additional care hours. We report that as a reading the data are consistent with, not a result the design identifies: mothers and fathers are not linked in the data, and the fathers’ coefficient is itself imprecise.

6.6 The extensive margin

Appendix Table A3 reports equation (1) with employment as the outcome. The DiD is an informative null: its standard error is half a percentage point and its pre-period coefficients pass their joint test. The employment DDD with the cell-based index is a null of a different kind. The smallest triple interaction the design could detect is about 2.0 percentage points per tenth of a unit of exposure, given the index’s standard deviation of 0.07. Against the US estimate of 0.78 percentage points per 10% rise in WFH (Harrington and Kahn, 2025) it is underpowered by a factor of about 2.6. The shortfall is structural, a binary outcome and a regressor with little dispersion, and re-clustering, combining the

Table 4: Hours DiD and DDD by age of the youngest child

Sample of mothers	DiD: Mother $\times$ Post			DDD: Mother $\times$ Post $\times$ WFH_Exposure			
	Coef.	SE	N	Coef.	SE	p (boot)	N
All mothers (Table 2)	0.228	0.181	251,857	3.224**	1.022	0.0558	246,326
Youngest child aged 0–4	0.127	0.211	158,424	3.656**	1.079	0.0551	154,499
Youngest child aged 5–9	0.162	0.253	130,782	3.211*	1.311	0.0944	127,419
Youngest child aged 10–14	0.159	0.278	123,280	2.741	1.525	0.0935	120,084
Youngest child aged 15–17	0.561	0.348	107,061	1.862	0.935	0.1020	104,079

Weekly hours, reference-week workers, standard controls; each row compares the named group of mothers with all childless women, so the rows are not independent samples. DiD clustered by individual; DDD clustered by occupation, calibrated exposure, with p (boot) the wild cluster bootstrap p -value of the triple interaction (9,999 Rademacher draws, null imposed). The first row repeats Table 2. No coefficient is dropped for collinearity in any column.

Table 5: Hours DiD and DDD by subgroup, and the fathers’ comparison

Subgroup	DiD: Mother $\times$ Post			DDD: Mother $\times$ Post $\times$ WFH_Exposure			
	Coef.	SE	N	Coef.	SE	p (boot)	N
All women (primary)	0.228	0.181	251,857	3.224**	1.022	0.0558	246,326
Jewish women	0.302	0.203	214,449	2.847**	0.913	0.0643	209,727
Arab women	−0.072	0.479	24,975	5.507*	2.674	0.1438	24,307
Men (fathers vs. childless)	−0.400*	0.188	261,543	−1.762	1.015	0.1494	251,779
95% CI, Jewish	[−0.095, 0.699]			[1.058, 4.635]			
95% CI, Arab	[−1.010, 0.866]			[0.266, 10.748]			
Jewish vs. Arab, z (p)	0.720 (0.471)			−0.942 (0.346)			
Women vs. men, z (p)	2.406 (0.016)			3.461 (0.001)			

Weekly hours, reference-week workers, standard controls. DiD clustered by individual; DDD clustered by occupation, with p (boot) the wild cluster bootstrap p -value of the triple interaction (9,999 Rademacher draws, null imposed). Men: “Mother” read as “Father,” a comparison population rather than a placebo (Section 5.5). Two-sample z -tests as described there. Confidence intervals use the normal critical value. The Arab-women DiD drops the religion category ‘other’, constant on that sample; the Arab-women DDD drops the religion category ‘other’, constant on that sample; no other column drops a coefficient.

two margins and refining the cell partition did not close it. The employment DDD is therefore *uninformative*, evidence neither for nor against a WFH-mediated employment effect, and the two nulls should not be read as one verdict.

7 Discussion

Among mothers who remain in work there is no post-2021 hours effect on average and a large one in teleworkable occupations. The triple interaction is about 3.2 hours per unit of exposure, and the implied change at zero exposure is if anything negative. This pattern is consistent with WFH access relaxing a binding time constraint for mothers in teleworkable jobs, the flexibility mechanism (Goldin, 2014). A general recovery common to all women would instead have raised hours across occupations, not only where the work can be done from home. The checks of the preceding section show that no single specification, occupation, sample or outcome coding produces the gradient. They are not independent confirmations: all share the headline’s forty occupation clusters, and the wild cluster bootstrap p -values beside them mostly lie above the 5% level (Tables 3–5). The gradient is identified off a concrete contrast: teaching and clerical support, which Israeli practice kept in person, placed low against the professional occupations placed high. Among the latter, science and engineering professionals (ISCO 21) carry the most weight; removing them gives the leave-one-out minimum of 2.415. ICT professionals, high on every exposure measure, work slightly against the gradient; removing them raises it to 3.287 (Appendix Figure A3).

The magnitude is economically meaningful where it applies. For a mother at the top quartile’s mean exposure of 0.59, the regression implies a post-2021 change of about 1.4 hours a week ($-0.54 + 3.22 \times 0.59$), close to the raw top-quartile difference-in-differences of Figure 1. That 1.4 is the fitted change *at* top-quartile exposure, distinct from the 1.71-hour fitted contrast *between* the quartile means. The relevant penalty is the one in those same jobs: column (2) of Table 2 puts the pre-period gap at that exposure at $-1.45 - 1.30 \times 0.59 \approx -2.2$ hours, so these mothers recovered roughly three fifths of it after 2021. At the bottom quartile’s mean exposure of 0.06 the implied change is -0.35 hours: mothers in the least teleworkable jobs recovered none of it. Commuting time alone does not account for the size. Workers save about 72 minutes per day worked from home and return roughly 40% of that time to their jobs, the rest going to leisure and care (Aksoy et al., 2023). Two remote days a week therefore add about one hour of paid work, not two and a half. The 1.4 hours also averages over *all* mothers in top-quartile jobs, yet even there only a minority of workers are at home in the reference week (Appendix Table A1). The response per mother who actually works from home is therefore several times larger than the commute she saves. That points to the channel of the framework in Section 2: flexibility in *when* hours are worked, hours split around care, rather than

travel time returned to work. The timing is consistent with a multi-year adjustment: the DDD event study rises from 2021 to 2023 (on the 2023 year, see Section 8).

A general recovery is not the only alternative to that mechanism. Three demand-side accounts also fit a gradient by teleworkability. First, employers in professional and ICT occupations may have changed the contracts they offer mothers after 2021, with no change in mothers' constraints. The saturated specification absorbs any occupation-by-year shock common to mothers and childless women, so this account needs a change aimed at mothers in teleworkable occupations specifically. Second, the 2021–22 technology hiring boom raised hours in exposed occupations for everyone. The fathers' opposite-signed gradient rules out a shock common to all workers in those occupations, but not one that fell on mothers alone. Third, usual hours are self-reported. If mothers working from home report the hours they are available while fathers report the hours they work, the two gradients would diverge as observed with no change in labor supplied. The child-age gradient does not separate the two readings. Teleworking mothers of young children report more interruptions to the workday (Pabilonia and Vernon, 2022). The gap between hours available and hours worked would then itself be largest for mothers of infants, the pattern the gradient shows. None of the three is excluded by the design. We read the exposure gradient as the time-constraint mechanism's signature rather than its proof.

Why hours and not employment? The employment DDD is not a failed test of the mechanism but an uninformative one (Section 6.6). The two results are coherent. The effect operates on *how much* mothers in teleworkable jobs work, not on *whether* they are employed, a leading channel of the child penalty (Kleven, Landaís and Sogaard, 2019). That is where our findings depart from the US evidence on employment and income (Harrington and Kahn, 2025). The policy reading is correspondingly narrow. WFH availability matters for how intensively mothers who already hold *teleworkable* jobs can work, not for whether mothers in general hold jobs. Access to those jobs is itself unequally distributed across mothers, so the population-wide gain is smaller than the gradient alone suggests and cannot be sized without representative weighting.

8 Limitations

Identification and inference. The triple difference's own pre-trend test (Section 6.2) shows no visible violation but lacks the power to rule one out. The exposure measure that delivers the headline is calibrated on post-period Israeli data, and the result requires reclassifying teaching. The fully pre-treatment external index yields a positive and insignificant estimate, and the checks of Section 6.3 bound how much of the gradient the reclassification carries. The sorting check finds no post-2021 movement of mothers across exposure levels, but the pre-period cell index bounds the effect only imprecisely. The headline's precision is the range of the three inference procedures in Section 6.2, not the

analytic p -value alone. The stars in the supporting rows of Tables 3–5 come from the analytic test, and the bootstrap p -values beside them mostly lie above 0.05. The permutation test assumes that the forty exposure scores are exchangeable across occupations under its null.

Exposure measurement. Exposure is assigned at the two-digit occupation level, forty groups, whereas sorting across firms, even within the same industry, is a first-order driver of Israeli gender gaps (Buzaglo-Baris, 2023). Assigning every woman her occupation’s mean teleworkability is Berkson-type error, which does not by itself bias the slope. Error in the occupation score itself attenuates the triple interaction toward zero. If mothers sort into more or less teleworkable jobs *within* an occupation, the bias can run in either direction. The calibrated score also mixes two scales (Appendix B). Occupation codes masked by the CBS for disclosure control are dropped when building exposure. A proxy check finds a small, statistically detectable difference in remote work between masked and unmasked rows (Appendix B).

The narrowed hours estimand. Hours are usual hours among women who were working, not among all women holding a job, and roughly 10% of each year’s employed sample is excluded on that basis. The exclusion is not innocuous for the triple interaction, because absence is related to exposure. Absence is highest in the least teleworkable quartile, and among mothers it fell after 2021 in the top quartile while rising in the bottom one (Appendix A). The bounds of Section 5.4 are therefore bounds on selection into the observed-hours sample. They handle only *excess* selection under monotonicity, and finding little to correct is not the same as the concern being absent. The absence margin can be checked directly, because employed women absent from work in the reference week still carry an occupation code. Adding them at their reported usual hours on 2018–2023, the years in which the survey recorded those hours, gives 3.076*** (SE 0.861; bootstrap $p = 0.07$) against the headline’s 3.224 (Table 3). The reference-week restriction is therefore not what produces the gradient. Trimming bounds on the absence margin itself are left for future work.

Sample, period and comparison group. The 2023 coefficient is the largest in both event studies, and the 2023 survey year includes the quarter in which the war that began on 7 October 2023 disrupted the labor market. Reserve call-ups bear on the fathers’ comparison, and school closures and evacuations bear on mothers’ hours. The estimate survives dropping the year, and 2021 and 2022 are individually significant on the triple difference. The 2023 point should nonetheless be held more loosely. Early 2021 likewise included a third lockdown, so the post-period is “post-acute” more cleanly in its later years than in its first. The comparison group includes women whose children are grown

and women who become mothers within the window, as in the US study (Harrington and Kahn, 2025). If part of it responds to the treatment as mothers do, the DiD is biased toward zero. Mothers and non-mothers are imbalanced on the categorical age code in a way that tracks exposure (Appendix Table A4). The age-interacted and reweighted specifications of Section 6.4 address this. The estimate stays positive in both but, after reweighting, falls below the design’s minimum detectable effect. Marital status is imbalanced in the same direction, and the two marital checks of that section leave the gradient positive but only marginally significant.

Survey weights. CBS survey weights are intentionally not applied in any outcome regression, so reported rates and coefficients are unweighted estimates on the analysis sample, not population-representative statistics. The one exception is internal to constructing the cell-based exposure index.

9 Conclusion

Using Israeli survey data, we ask whether the expansion of remote work coincided with a change in the motherhood penalty in usual weekly hours. The average change among women observed working is small, whereas the post-2021 change in the mother/comparison gap increases with an occupation’s calibrated WFH exposure. The strongest contrast is at the top of the exposure distribution and among mothers with younger children. Those patterns are consistent with greater scheduling flexibility helping some already employed mothers supply more paid hours.

The result is conditional and does not establish a universal benefit from remote work. The exposure score partly uses post-period WFH information; the wholly external score yields a weaker result, pre-trend tests have limited power, and inference varies with the procedure. Employment is an important separate margin, but its exposure interaction is too imprecise here to settle the question. The data also contain no earnings amount. Stronger evidence would link hours with earnings and household time use, measure exposure before treatment in the Israeli setting, and estimate women’s and men’s responses together on a design that better separates access to WFH from occupation choice.

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A Data and sample details

Hours coding. The CBS records usual weekly hours in bins. Each bin is assigned its median (4, 11, 18, 25.5, 32, 37, 42, 47, 54.5 and 78.5 hours). The two irregular-hours codes are imputed from observed medians within the same period. Mid-distribution bins are about five hours wide, so the headline estimate, smaller than one bin width, is best read as a shift in the share of women crossing a bin boundary. The 2017 file also codes reference-week absence differently. Employed respondents absent from work that week were recorded at zero usual hours in 2017 and at a real value from 2018 onward. Left uncorrected, this places roughly 5,200 spurious zeros in the pre-period. Because absence is more common among mothers (12.4% of employed mothers against 7.3% of childless women), the zeros fall unevenly across the two groups. Defining hours on reference-week workers in *every* year removes them and makes the hours population identical across years, at a cost of about 10% of each year’s employed sample.

Reference-week absence and exposure. Absence is related to exposure. Among employed women with a matched occupation, the absence shares by the quartiles of Figure 1 are 14.4%, 8.5%, 9.5% and 9.6%, highest in the least teleworkable quartile. After 2021 mothers’ top-quartile share fell from 12.1% to 10.4% and their bottom-quartile share rose from 15.5% to 16.9%, while childless women’s top-quartile share is 6.1% in both periods. Selection into the observed-hours sample therefore includes movement across the absence margin, which is why the bounds of Appendix C are defined on that sample.

Panel structure. The survey is a rotating panel. The 258,175 reference-week workers of Table 1 come from 64,602 distinct respondents, about four observations each and two to three within a single survey year. Repeat respondents account for 96% of rows. The regressions use 251,857 of those rows from 63,199 respondents; the 6,318 dropped rows have a missing education code, the only control with missing values. Treating the observations as independent would understate standard errors by a factor of roughly 1.8, hence the individual clustering of every DiD and every descriptive interval. The panel does not support within-person identification. Of the 80,560 women in the full sample, 1.9% are interviewed on both sides of the pre/post boundary, too few for an individual fixed-effects design.

Hours year by year. Panel (a) of Figure A1 unpools the pre-period. The raw mother-minus-non-mother gap widens monotonically across the three pre-period years, by 0.3 hours in all, but each year lies inside the others’ confidence bounds. The post-period path is neither monotone nor individually distinguishable. The timing reading in the Discussion therefore rests on the event studies of Figure 2, not on this series.

Commuting. Mothers were the *more* mobile group in every year. In 2019 they worked outside their locality of residence 48.2% of the time, against 46.1% of childless women. Their commuting also fell less in 2021: 1.5 against 3.9 percentage points, a difference of 2.39 points (SE 1.17). This is consistent with remote work relaxing a time constraint for women who were already commuting. It is descriptive evidence only, and no specification uses locality of work as an outcome.

B Exposure construction

The external score. The Dingel–Neiman teleworkability score (Dingel and Neiman, 2020) reaches the two-digit ISCO-08 level through the BLS crosswalk between ISCO-08 and the 2010 SOC (Bureau of Labor Statistics, 2012). Each O*NET-SOC occupation in their classification is matched to every ISCO-08 unit group the crosswalk lists for its six-digit SOC code. The score of a sub-major group is the unweighted mean of their binary indicator over the matched pairs. Armed-forces groups have no counterpart in their data and carry no score. The replication code re-derives the forty scores (Table A1) from the two public source files. Teaching (ISCO 23) is the clearest case of a US task-based score misclassifying Israeli practice. It scores 0.966 on the external index, yet Israeli schools, closed and reopened in stages through 2020 and early 2021 (Somekh et al., 2021; Hale et al., 2021), were fully in person throughout the 2022–23 survey years. Teachers did not work from home.

The calibration rule. The external score is replaced by the realized 2022–23 Israeli WFH share only where two conditions hold. The gap between the two must exceed 0.5, and a one-sided test at the 95% level, with standard errors clustered by individual, must distinguish it from sampling noise. Ten of the forty two-digit occupations are swapped, among them teaching, clerical support (ISCO 41) and ICT professionals (ISCO 25). Teaching falls from 0.966 to 0.069. The realized shares pool men and women; Table 3 re-derives them from men alone. The realized index reported as a bound is the realized Israeli WFH share by occupation, anchored to 2021 because no 2020 extract exists, with a minimum of 200 observations per occupation.

Masked occupation codes. The CBS masks some two-digit occupation codes for disclosure control. Those rows carry no exposure score and drop out of the hours DDD. A partially masked code still identifies the one-digit major group, so a proxy check compares realized WFH between masked and unmasked employed women within the same major group in 2022–23. Among professionals (major group 2), masked rows work from home at a share of 0.29 against 0.16 for unmasked ones, on 207 masked observations. In the other major groups the masked rows are fewer and the gaps are small or negative. Pooled across major groups, with major-group fixed effects and standard errors clustered by individual, the masked-minus-unmasked difference in the share working from home is 0.045 (SE 0.021) on 85,454 employed women. The difference is significant at the 5% level and negligible in explanatory power (within R^2 below 0.0001). The dropped rows are therefore somewhat more teleworkable than those the index is built from: a small difference on few rows, not a clean bill of health.

Fit. Across the forty occupations, on men and women pooled over 2021–23, the calibrated score has a correlation of 0.53 with the share who worked from home in the reference week. Its occupation-size-weighted slope on that share is 0.40 (SE 0.09), so the score is relevant. The raw external index fits better in levels (correlation 0.74), because swapping ten occupations onto realized shares while leaving thirty on task-based probabilities mixes two scales. The calibration’s contribution is not a better cross-sectional fit but the removal of ten grossly mis-scored occupations from the treated tail. Table A1 lists the external, realized and calibrated scores by occupation; panel (b) of Figure A1 plots the realized reference-week share against the calibrated score.

Sensitivity to the swap threshold. Table A2 refits the hours DDD with the calibration’s swap threshold moved from 0.30 to 0.90. Lowering it swaps more occupations and the estimate grows noisier. From 0.75 upward only teaching and clerical support are swapped. At 0.90, above the largest gap, nothing is swapped and the row is the external-index estimate of Table 3. The estimate is positive at every threshold. The step from the external index to any positive threshold is the reclassification of teaching, which Table 3’s teaching-only row isolates.

C The selection correction

Let $s_{ab} = \Pr(\text{hours observed} \mid \text{Mother} = a, \text{Post} = b)$, the probability of being employed and working in the reference week. Under parallel trends in selection the counterfactual for post-period mothers is $s_{11}^* = s_{10} + s_{01} - s_{00}$. If s_{11} exceeds it, the excess share $1 - s_{11}^*/s_{11}$ is trimmed from the (Mother = 1, Post = 1) hours distribution, from the top for the lower bound and from the bottom for the upper bound, and the model is re-estimated on each

Table A1: External, realized and calibrated WFH exposure by two-digit occupation

ISCO	Occupation (ISCO-08 sub-major group)	Ext.	Real. 22–23	Swap	Calib.	Ref.-wk 21–23	<i>N</i>
11	Chief executives and legislators	0.900	0.155	yes	0.155	0.320	2,855
12	Administrative and commercial managers	0.892	0.143	yes	0.143	0.336	3,859
13	Production and specialised services managers	0.638	0.097	yes	0.097	0.199	9,990
14	Hospitality, retail and other services managers	0.773	0.066	yes	0.066	0.132	2,827
21	Science and engineering professionals	0.644	0.210		0.644	0.349	13,228
22	Health professionals	0.101	0.059		0.101	0.088	9,665
23	Teaching professionals	0.966	0.069	yes	0.069	0.184	20,496
24	Business and administration professionals	0.943	0.233	yes	0.233	0.449	10,479
25	ICT professionals	1.000	0.360	yes	0.360	0.615	14,638
26	Legal, social and cultural professionals	0.678	0.215		0.678	0.371	13,344
31	Science and engineering associate professionals	0.144	0.059		0.144	0.106	3,926
32	Health associate professionals	0.057	0.080		0.057	0.084	2,203
33	Business and administration associate professionals	0.684	0.123	yes	0.123	0.223	22,415
34	Legal, social and cultural associate professionals	0.577	0.142		0.577	0.206	5,480
35	ICT technicians	0.750	0.244		0.750	0.380	2,491
41	General and keyboard clerks	1.000	0.098	yes	0.098	0.144	4,846
42	Customer services clerks	0.292	0.096		0.292	0.159	3,875
43	Numerical and material recording clerks	0.526	0.071		0.526	0.152	4,948
44	Other clerical support workers	0.625	0.083	yes	0.083	0.161	1,049
51	Personal service workers	0.271	0.146		0.271	0.144	7,808
52	Sales workers	0.200	0.050		0.200	0.063	8,704
53	Personal care workers	0.300	0.046		0.300	0.045	17,206
54	Protective services workers	0.100	0.008		0.100	0.011	3,969
61	Market-oriented skilled agricultural workers	0.182	0.027		0.182	0.051	1,325
62	Market-oriented skilled forestry, fishery and hunting workers	0.179	0.000		0.179	0.045	22
63	Subsistence farmers, fishers, hunters and gatherers	0.000	0.375		0.000	0.375	8
71	Building and related trades workers	0.027	0.008		0.027	0.016	5,208
72	Metal, machinery and related trades workers	0.000	0.008		0.000	0.013	3,982
73	Handicraft and printing workers	0.138	0.118		0.138	0.129	666
74	Electrical and electronic trades workers	0.000	0.022		0.000	0.042	3,424
75	Food, wood, garment and other craft workers	0.073	0.122		0.073	0.109	2,304
81	Stationary plant and machine operators	0.000	0.019		0.000	0.017	2,407
82	Assemblers	0.000	0.012		0.000	0.015	1,166
83	Drivers and mobile plant operators	0.237	0.006		0.237	0.006	8,480
91	Cleaners and helpers	0.000	0.004		0.000	0.003	3,474
92	Agricultural, forestry and fishery labourers	0.000	0.007		0.000	0.016	192
93	Labourers in mining, construction, manufacturing and transport	0.120	0.006		0.120	0.008	2,659
94	Food preparation assistants	0.000	0.013		0.000	0.011	649
95	Street and related sales and service workers	0.000	0.000		0.000	0.043	23
96	Refuse workers and other elementary workers	0.250	0.022		0.250	0.024	493

Ext.: the Dingel–Neiman teleworkability score (Dingel and Neiman, 2020) at the two-digit ISCO-08 level, via the crosswalk described above. Real. 22–23: share of employed respondents (men and women aged 25–59) who usually work from home in 2022–23, the calibration’s benchmark; Swap: the calibration replaced the external score with it. Calib.: the score the hours DDD uses. Ref.-wk 21–23: share who worked from home in the survey reference week over 2021–23, men and women pooled, with *N* the pooled employed sample carrying that item. Codes 1–3 (armed forces) carry no external score and are not in the sample.

Table A2: Hours DDD by calibration gap threshold

Gap threshold	Occupations swapped	Coefficient	SE	p	Clusters
0.30	15	3.930	2.819	0.1712	40
0.35	15	3.930	2.819	0.1712	40
0.40	15	3.930	2.819	0.1712	40
0.45	12	4.239***	1.177	0.0009	40
0.50	10	3.224**	1.022	0.0031	40
0.55	8	3.096**	0.939	0.0021	40
0.60	7	2.902***	0.729	0.0003	40
0.65	6	2.475***	0.673	0.0007	40
0.70	4	2.657***	0.565	< 0.0001	40
0.75	2	2.319***	0.504	< 0.0001	40
0.80	2	2.319***	0.504	< 0.0001	40
0.85	2	2.319***	0.504	< 0.0001	40
0.90	0	0.672	0.884	0.4517	40

Mother $\times$ Post $\times$ WFH_Exposure from equation (2), women working in the reference week, standard controls, clustered by occupation, with the exposure index recalibrated at the gap threshold in the first column (the significance test on the gap is unchanged). The 0.50 row is the headline of Table 2; a row with no occupation swapped is the external Dingel–Neiman index. Stars follow the paper’s legend on the analytic $t(G - 1)$ test.

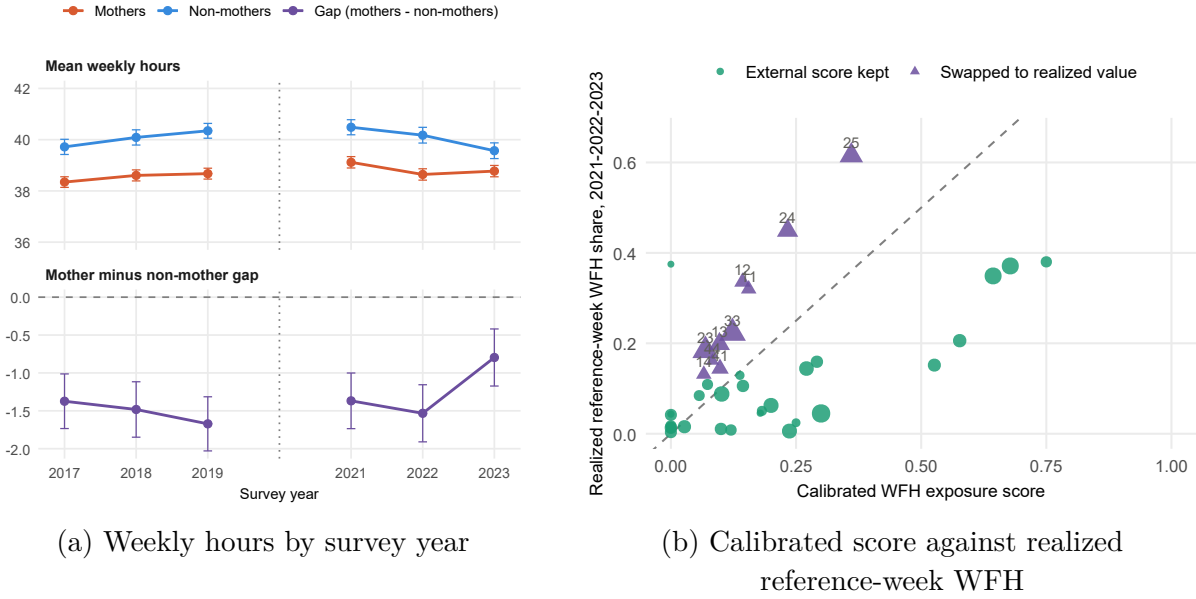


Figure A1: Weekly hours by survey year, and the exposure first stage
Notes: Panel (a): women aged 25–59 who worked in the reference week, unweighted means with no controls; the upper panel plots each group’s mean weekly hours on an axis spanning 36 to 42 hours, the lower panel their difference, with 95% confidence intervals clustered by individual; the dotted rule marks the excluded 2020 survey year, and the series break there. Panel (b): one point per two-digit occupation, sized by the pooled employed sample in 2021–23; triangles are the ten occupations the calibration swapped to their realized value, labelled by ISCO-08 code; the dashed line is the 45-degree line. The swapped occupations sit above it because their calibrated value is the usual-place share, which is lower than the reference-week share everywhere; the thirty unswapped ones lie below it, the visual form of the scale-mixing described above.

trimmed sample. Trimming one tail assumes monotone selection, and the construction handles only excess selection in that cell. For the DDD the counterfactual and trim are computed within each quartile of the cell-based exposure index, which assumes selection depends on exposure only through that quartile, and equation (2) is refit on the recombined sample with the occupation-level regressor and occupation clustering. Because the parameter is partially identified, the Imbens–Manski interval for the identified set is reported with the bounds (Imbens and Manski, 2004).

For the DiD, $s_{11} = 0.696$ against $s_{11}^* = 0.683$, and trimming the 1.9% excess gives bounds of -0.517 (SE 0.178) and 0.802 (SE 0.179) around the 0.228 point estimate, with Imbens–Manski 95% interval $[-0.810, 1.096]$. For the DDD the excess is present in all four quartiles but modest, a trim of 0.80% to 4.48%, 1,396 of 72,820 post-period working mothers; the bounds are 3.126 (SE 1.144) and 3.443 (SE 1.022) around an untrimmed 3.232 (SE 1.054), which differs from Table 2’s 3.224 because this sample additionally requires a matched cell-based exposure, and the Imbens–Manski interval is $[1.021, 5.324]$.

D Additional results

Figure A2 plots the reference distribution of the permutation test: under the sharp null the central 95% of cluster-robust t -statistics lie between -2.96 and 2.60 , and the observed 3.15 is exceeded in absolute value by 31 of 999 draws. Table A3 reports the extensive-margin DiD and DDD discussed in Section 6.6. Table A4 reports the pre-period covariate balance between mothers and childless women within each exposure quartile of the hours DDD’s estimation sample, and Figure A3 the forty leave-one-occupation-out refits of the headline estimate.

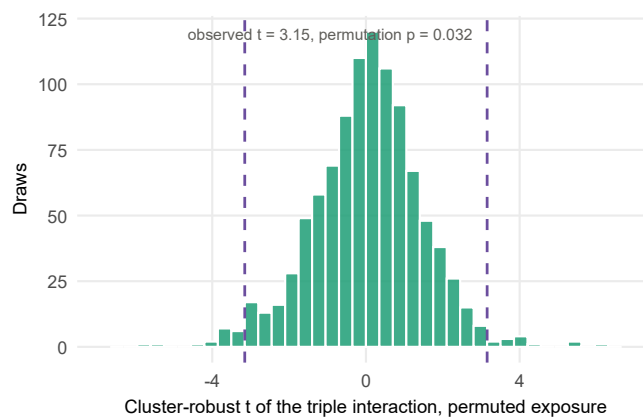


Figure A2: Permutation distribution of the hours DDD triple interaction

Notes: Histogram of the cluster-robust t -statistic of $\text{Mother} \times \text{Post} \times \text{WFH_Exposure}$ over 999 random reassignments of the forty exposure scores across occupations, with the same sample, controls and clustering as column (2) of Table 2. Dashed lines mark $\pm$ the observed statistic (3.15); $p = 0.032$ counts the observed draw (Phipson and Smyth, 2010).

Table A3: Extensive margin: employment DiD and DDD

	(1) DiD	(2) DDD, cell-based exposure
Mother $\times$ Post $\times$ WFH_Exposure		−0.0257 (0.0725)
Mother $\times$ Post	−0.0053 (0.0052)	0.0065 (0.0168)
Mother $\times$ WFH_Exposure		−0.1404 (0.0803)
Post $\times$ WFH_Exposure		−0.1555** (0.0566)
WFH_Exposure		0.0131 (0.0565)
Mother	−0.0034 (0.0041)	
Post	0.0153*** (0.0043)	
Constant	0.4926*** (0.0085)	
MDE ($\alpha = 0.05$, power 0.80), per unit		0.2031
MDE per SD of exposure		0.0143
as % of baseline employment rate		1.85%
Observations	364,784	355,984
R^2	0.227	0.221
Clustering	individual	demographic cell (≈ 210)

Dependent variable: employment indicator; full sample of employed and non-employed women; standard controls in both columns, four decimals because the outcome is a probability. Column (2) adds Mother $\times$ age-group interactions, so its Mother and Post main effects are not separately reported, and uses the demographic-cell index of Section 3.2, whose standard deviation on this sample is 0.0706; the per-SD MDE row is the interpretable one. Baseline employment rate 0.7737. No coefficient is dropped for collinearity in any column.



Figure A3: Leave-one-occupation-out: the hours DDD triple interaction with each occupation removed

Notes: Each point is $\text{Mother} \times \text{Post} \times \text{WFH_Exposure}$ from equation (2) re-estimated without the occupation named on the axis (column (2) of Table 2 on 39 occupations), with its 95% confidence interval clustered by occupation; rows are sorted by the estimate. The dashed line is the headline estimate on all forty occupations and the shaded band one headline standard error either side of it; the dotted line marks zero. Triangles would mark refits outside the band; there are none.

Table A4: Pre-period covariate balance by quartile of occupation-level WFH exposure

Mothers minus childless women, pre-period	Q1	Q2	Q3	Q4
Age-group code (CBS 3–7), mean	−0.54*** (0.03)	−0.44*** (0.03)	−0.33*** (0.03)	0.08* (0.04)
Married (%)	32.2*** (1.1)	33.7*** (1.1)	41.5*** (0.9)	43.1*** (1.3)
Academic degree (%)	10.8*** (1.2)	4.5*** (1.2)	1.5 (1.0)	1.8 (1.2)
Below high school (%)	−3.2*** (0.6)	−1.7*** (0.4)	−3.3*** (0.5)	−0.2 (0.2)
Jewish household head (%)	1.6 (1.0)	0.5 (0.7)	14.3*** (0.9)	3.1*** (0.8)
Center or Tel Aviv district (%)	−4.4*** (1.2)	−3.4** (1.2)	−8.8*** (1.1)	−4.5** (1.4)
Arab (%)	1.9* (0.9)	0.2 (0.6)	3.7*** (0.6)	−0.9 (0.6)
Mothers	22,976	24,664	22,562	16,377
Childless women	11,420	11,586	17,244	8,352
Occupations	15	7	11	6
Exposure range	0.00–0.10	0.10–0.14	0.14–0.30	0.36–0.75

Women working in the reference week in 2017–2019 with a matched two-digit occupation, the hours DDD’s pre-period estimation sample, split into the exposure quartiles of Figure 1. Each cell is the mother-minus-childless difference in the row variable within the quartile, with a standard error clustered by individual in parentheses; shares are in percentage points, the age-group code in code units (CBS codes 3–7 span ages 25–59 in five-year bands). Stars follow the paper’s legend on a normal reference. Unweighted.